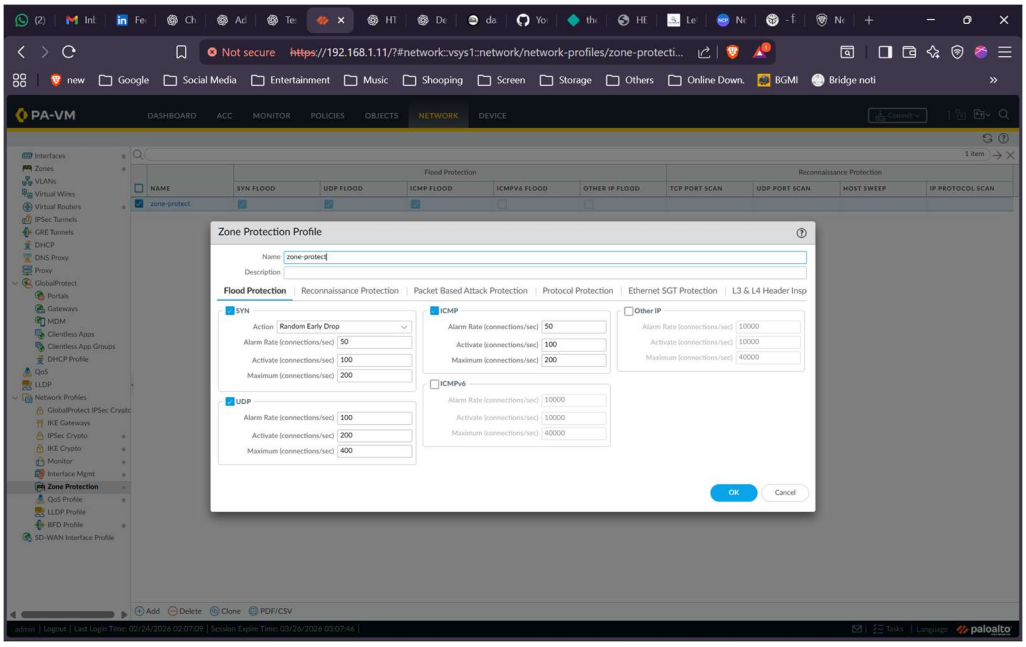
Palo Alto DoS Protection Configuration & Validation Report

This report presents the implementation and validation of Zone Protection and Denial-of-Service (DoS) Protection mechanisms on Palo Alto PA-VM firewall. The objective of this configuration is to safeguard internal network resources from flood-based attacks originating from untrusted networks.

Testing was performed using an Ubuntu virtual machine to simulate attack traffic. Validation was confirmed through firewall threat logs and monitoring tools.

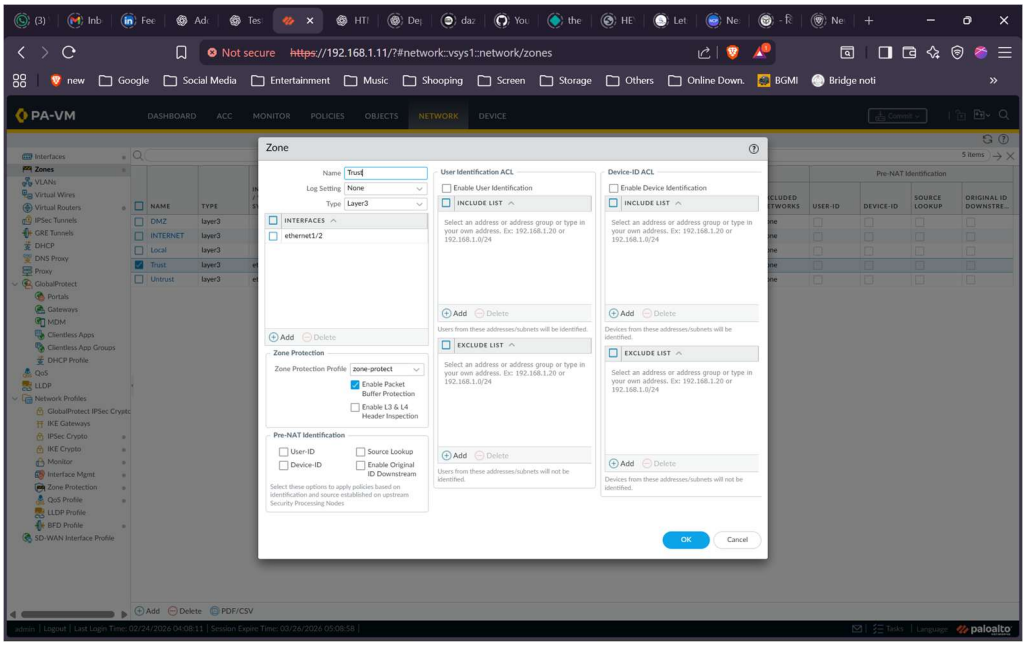
1. Zone Protection Profile – Flood Protection Configuration

A Zone Protection profile was created to protect the firewall at the zone level. Flood protection mechanisms including SYN Flood, UDP Flood, ICMP Flood, and other IP flood types were configured with defined alarm, activate, and maximum thresholds. These thresholds ensure early detection and mitigation of abnormal traffic spikes before resource exhaustion occurs.



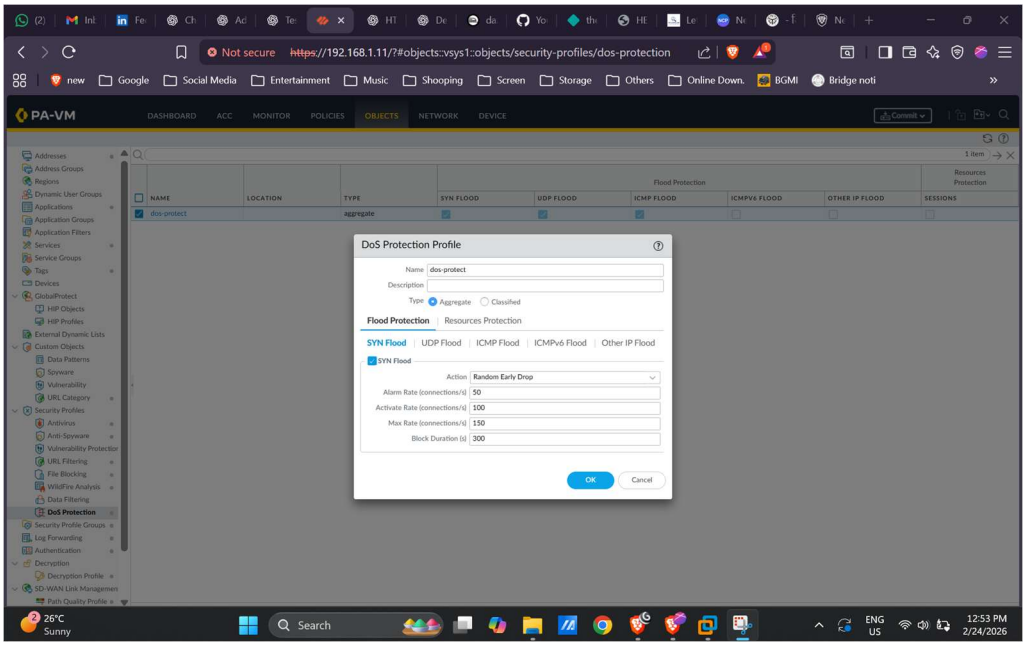
2. Zone Configuration – Applying Protection Profile

The configured Zone Protection profile was applied to the Trust zone. This ensures that all traffic entering or leaving the zone is subject to flood protection checks. Buffer protection and packet-based attack protection settings were also enabled where required.



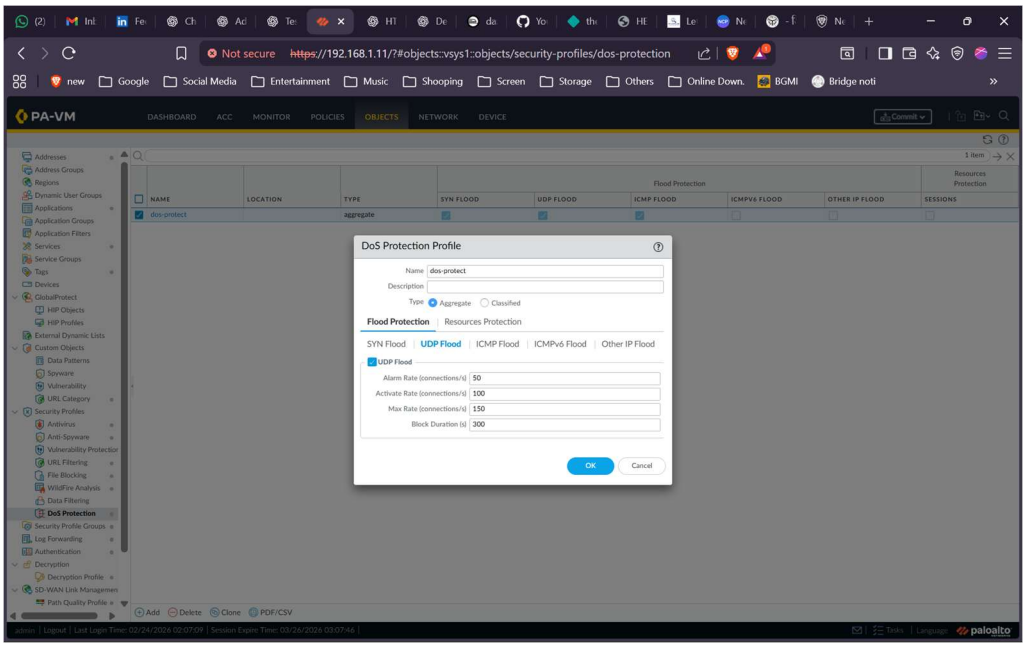
3. DoS Protection Profile – SYN Flood Thresholds

An Aggregate DoS Protection profile was created to protect specific servers. SYN flood protection thresholds were defined with alarm rate, activation rate, maximum rate, and block duration. The firewall is configured to perform Random Early Drop when thresholds are exceeded.



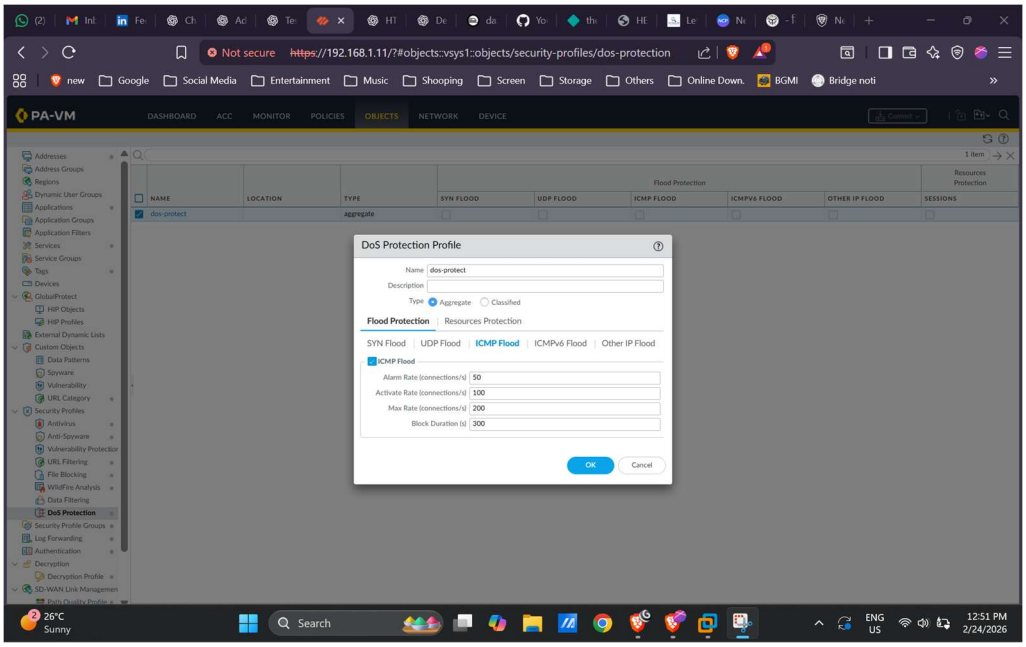
4. DoS Protection Profile – UDP Flood Configuration

UDP flood protection parameters were configured to detect high-rate UDP packet transmission. This prevents attackers from overwhelming services that rely on UDP-based communication.



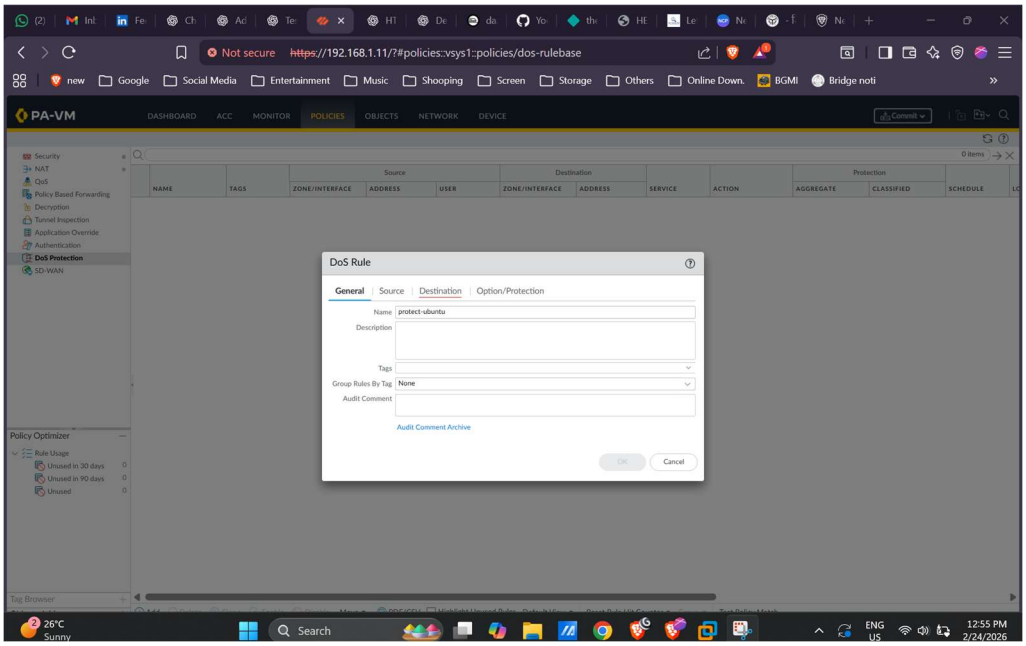
5. DoS Protection Profile – ICMP Flood Configuration

ICMP flood thresholds were configured to mitigate ping flood attacks. When ICMP packet rates exceed configured values, the firewall drops excessive traffic and logs the event with critical severity.



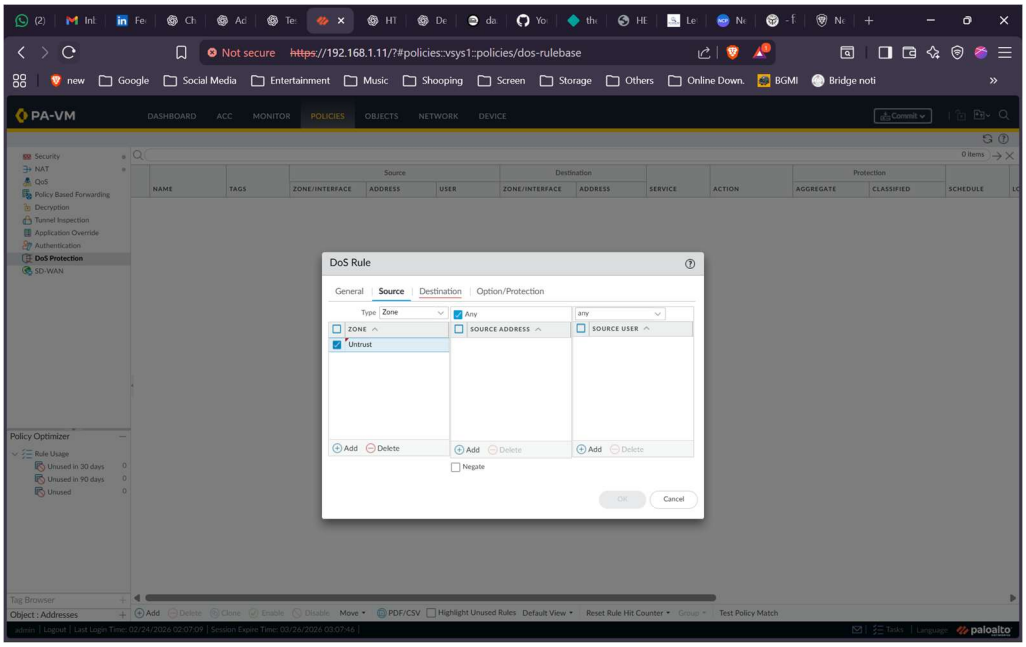
6. DoS Policy Rule – General Configuration

A DoS rule was created to specifically protect the internal Ubuntu server. This rule defines the traffic scope and associates the Aggregate DoS Protection profile to enforce threshold-based protection.



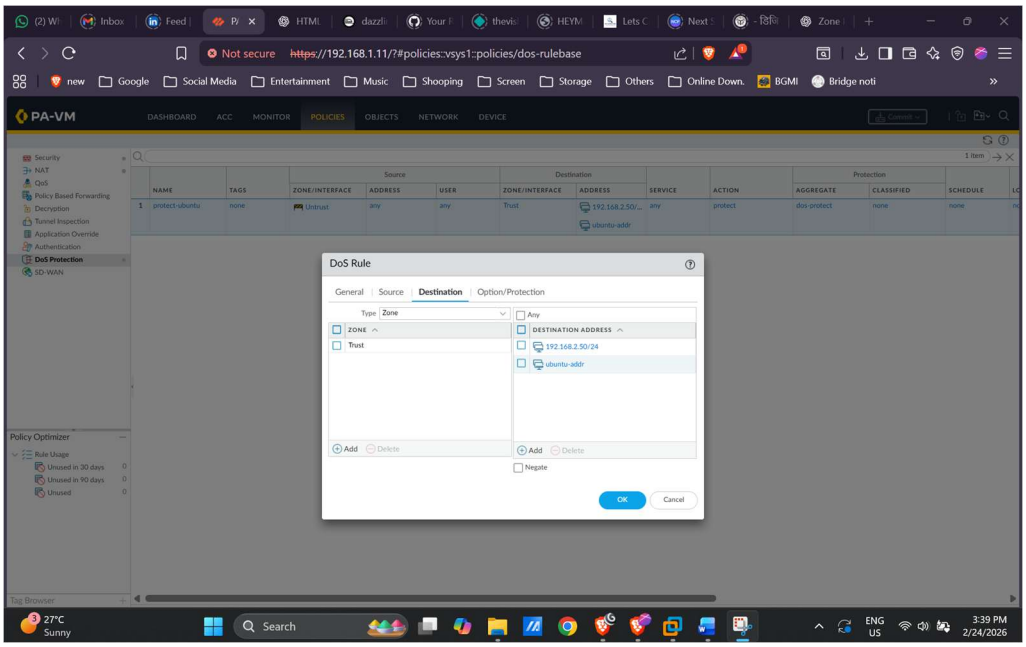
7. DoS Policy Rule – Source Configuration (Untrust Zone)

The source zone was configured as 'Untrust', representing external or potentially malicious traffic. This ensures that incoming traffic from external networks is inspected.



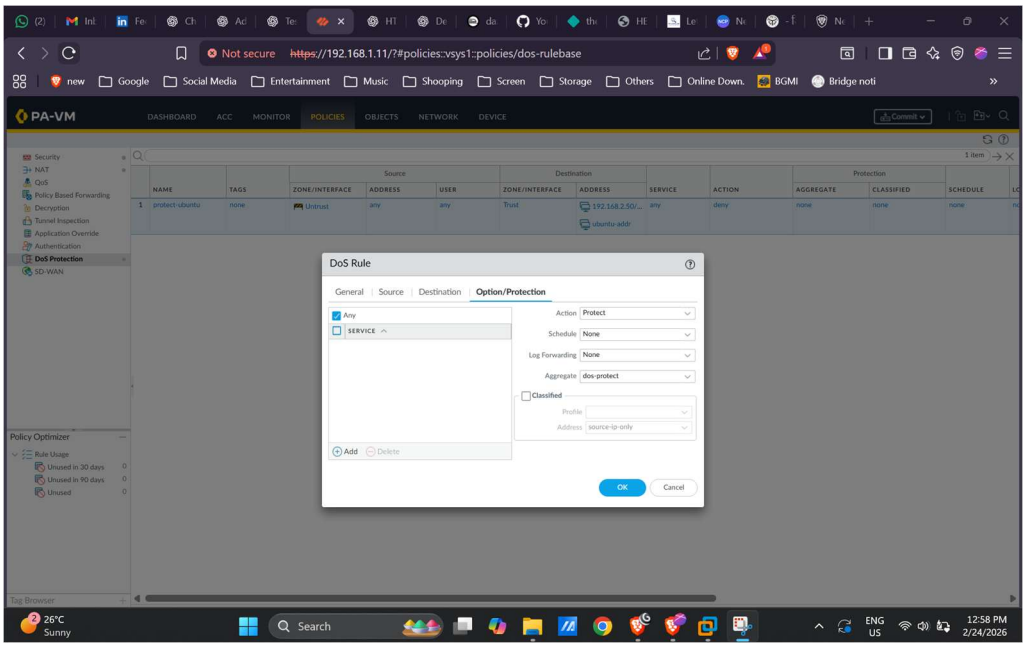
8. DoS Policy Rule – Destination Configuration (Protected Server)

The destination zone was set to 'Trust' and the internal Ubuntu server IP address was selected. This restricts protection specifically to the critical internal resource being tested.



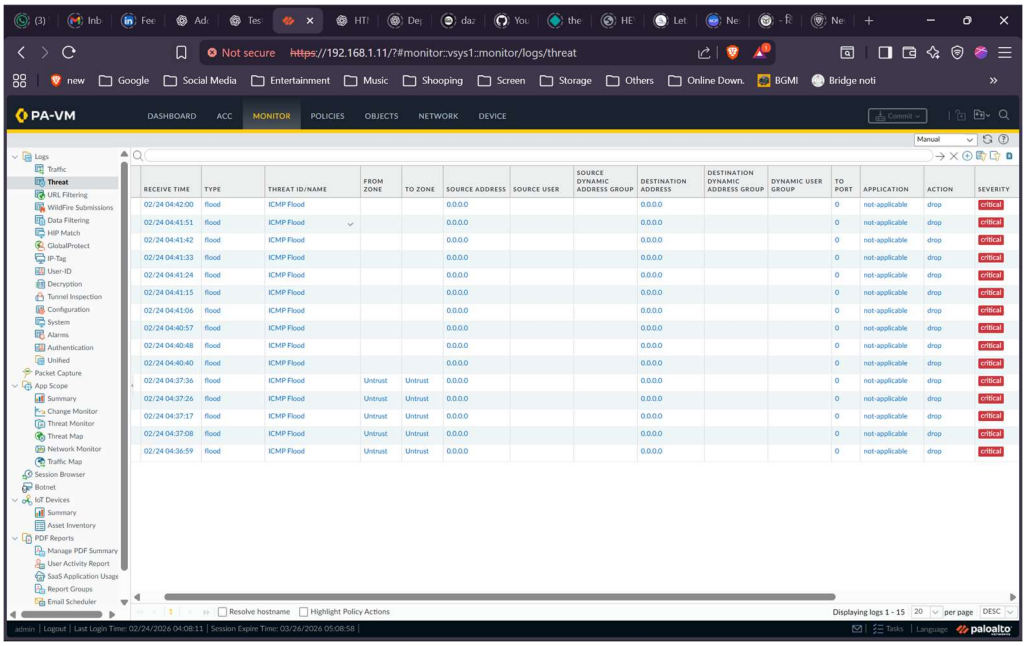
9. DoS Rule – Protection Profile Application

Under the Option/Protection tab, the Aggregate DoS Protection profile was applied with action set to 'Protect'. This activates the configured flood thresholds for matching traffic.



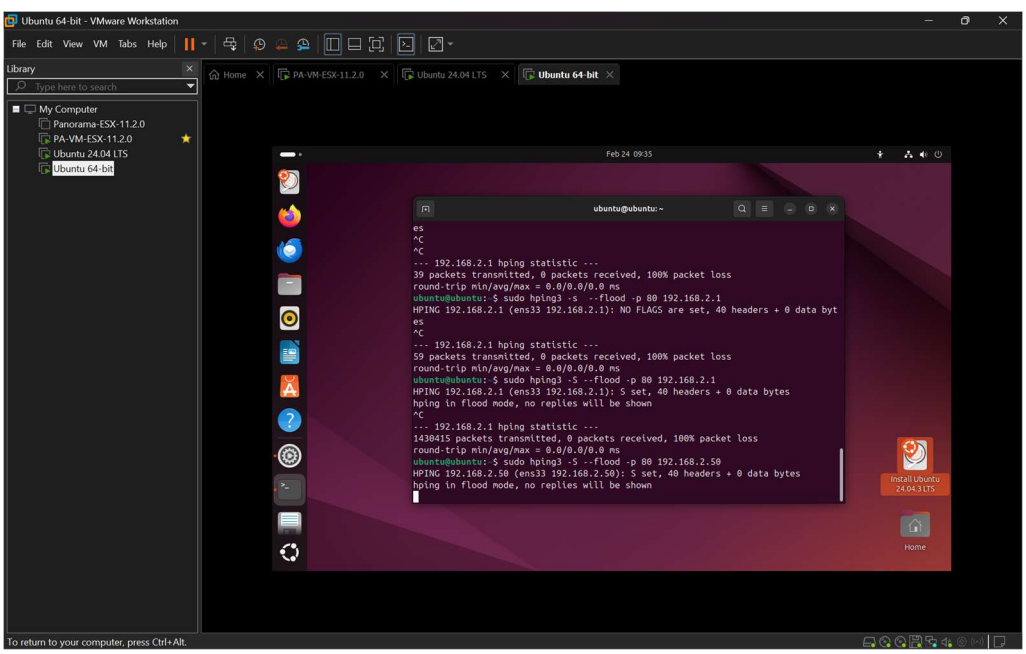
10. Monitoring and Validation – Threat Logs

Threat logs were monitored under the Monitor tab. The logs clearly indicate ICMP flood detections with action set to 'drop' and severity marked as 'critical'. This confirms successful detection and mitigation.



11. Attack Simulation – Ubuntu (hping3 Testing)

The hping3 tool was used from the Ubuntu virtual machine to generate ICMP and TCP flood traffic. Flood mode (-flood) was used to rapidly transmit packets toward the protected server. The absence of replies and corresponding firewall logs confirm that mitigation was successful.



Conclusion

The Palo Alto PA-VM firewall was successfully configured with both Zone Protection and Aggregate DoS Protection profiles to defend against flood-based attacks. Threshold-based controls were implemented for SYN, UDP, and ICMP traffic to detect abnormal packet rates and trigger mitigation actions such as Random Early Drop and traffic blocking.

A DoS policy rule was applied to protect a specific internal Ubuntu server from traffic originating in the Untrust zone. Attack simulations were conducted using the *hping3* tool to generate high-rate ICMP and TCP flood traffic. Monitoring results in the Threat Logs confirmed that the firewall accurately detected the flood attempts, dropped malicious packets, and classified the events with critical severity.

Overall, the configuration effectively demonstrated proactive detection and mitigation of DoS attacks, ensuring protection of internal network resources and validating the proper functioning of Palo Alto's security mechanisms.

— Vishesh Dixit